Explicit rational counterexamples to Simon's Le Roy conjecture

A beta-product density and a certified complete-monotonicity boundary

Keio Research Notes*

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Abstract

For the Le Roy function $F^{(\gamma)}_{\alpha,\beta}(z) = \sum_{n\geq 0} z^n/\Gamma(\beta+\alpha n)^\gamma$, we give a completely specified counterexample to Simon's proposed equivalence between complete monotonicity and logarithmic infinite divisibility. The function $F^{(2)}_{1/3,3/7}(-x)$ is completely monotone, whereas its unique representing probability law is not logarithmically infinitely divisible. A beta-product integral proves positivity of the density for $x^3 \geq 54$; an exact rational Bernstein certificate, with three rigorously enclosed gamma constants, proves positivity on $[0,4]$. The failure of the proposed necessary condition reduces to $21^7 < 2^{31}$. We also enclose the complete-monotonicity threshold at $(\alpha,\gamma) = (1/3,2)$ in the interval $(0.428292245841, 0.428292245842)$, prove that its density has exactly one zero, of order two, obtain an explicit interval of counterexamples, and prove a 3/2-power law for the negative mass born below the boundary. All certificates are included in the single source file. The qualitative refutation was already present in corpus item [116]; the contributions here are an explicit independent proof, a rigorous threshold enclosure, and uniqueness of the critical contact.

1 The external conjecture and the precise result

A smooth function $f : (0,\infty) \to \mathbb{R}$ is completely monotone (CM) when $(-1)^n f^{(n)}(x) \geq 0$ for every integer $n \geq 0$ and every $x > 0$. A positive random variable is logarithmically infinitely divisible (log-ID) when its logarithm has an infinitely divisible distribution on $\mathbb{R}$.

Simon [1, Section 3] proposed the following equivalence for all positive α, β, γ:

$$F^{(\gamma)}_{\alpha,\beta}(-x) \text{ is CM} \quad \Longleftrightarrow \quad z + \gamma(z^{\beta-\alpha} - z^\beta) \leq 1 \quad (0 < z < 1). \tag{1.1}$$

The scalar condition gives a nonnegative Levy kernel and is sufficient. Its claimed necessity is the part refuted here. The same paper explains the equivalent formulation in terms of log-ID of the representing law.

Throughout this manuscript, $\alpha = 1/3$ and $\gamma = 2$. Put

$$g_\beta(x) = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!\,\Gamma(\beta-(n+1)/3)^2}, \qquad q_\beta(x) = \Gamma(\beta)^2 g_\beta(x). \tag{1.2}$$

We prove below that q_β is an integrable signed density of mass one, with the desired Mellin transform. The series is entire in x; no positivity is assumed in its definition.

*AI-assisted mathematical research manuscript. Analytical arguments and executable certificates were checked internally; this is not external peer review or a claim of historical priority.

Theorem 1.1 (A small rational counterexample)**.** *There exists a positive random variable X, with a strictly positive density on $(0,\infty)$, such that*

$$\mathbb{E}X^s = \frac{\Gamma(1+s)\Gamma(3/7)^2}{\Gamma(3/7+s/3)^2} \quad (\operatorname{Re} s > -1), \qquad \mathbb{E}e^{wX} = \Gamma(3/7)^2 F^{(2)}_{1/3,3/7}(w) \quad (w \in \mathbb{C}).$$

Consequently $F^{(2)}_{1/3,3/7}(-x)$ is CM. However, $\log X$ is not infinitely divisible, and the right-hand side of (1.1) fails at $z = 1/64$.

Define the exact rational numbers

$$b_{\mathrm{L}} = \frac{428292245841}{10^{12}}, \qquad b_{\mathrm{U}} = \frac{428292245842}{10^{12}}. \tag{1.3}$$

Theorem 1.2 (Boundary enclosure and a unique double contact)**.** *There is a unique number $\beta_* \in (b_{\mathrm{L}}, b_{\mathrm{U}})$ for which*

$$F^{(2)}_{1/3,\beta}(-x) \text{ is CM} \quad \Longleftrightarrow \quad \beta \geq \beta_*, \qquad \beta > 0.$$

The critical density q_{β_*} has exactly one zero on $[0,\infty)$. This zero x_* belongs to $(1/2, 53/100)$, and

$$q_{\beta_*}(x_*) = q'_{\beta_*}(x_*) = 0, \qquad q''_{\beta_*}(x_*) > \frac{\Gamma(\beta_*)^2}{20}.$$

In particular, its order is exactly two. The equations $g_\beta(x) = \partial_x g_\beta(x) = 0$ have exactly one solution in $[b_{\mathrm{L}}, b_{\mathrm{U}}] \times [1/2, 53/100]$. For every $\beta \in [b_{\mathrm{U}}, 13/30]$, the function is CM and its representing law is not log-ID.

What is new relative to the supplied papers. Corpus item [116] [4] already proves a qualitative strict separation for all $\gamma > 1$, $0 < \alpha < 1/\gamma$, without specifying a rational counterexample. The preceding equivalence paper [5] gives, among other results, the lower bound $\beta_* > b_{\mathrm{L}}$. This manuscript does not count the same external conjecture twice. Its additional content is a short explicit rational triple, an independent beta-product proof of its global positivity, the matching upper bound $\beta_* < b_{\mathrm{U}}$, and the uniqueness and quadratic nature of the boundary contact. Neither the qualitative strict-gap theorem nor unproved uniform saddle estimates are used in the proof below.

Proof architecture. An exact beta-product identity handles the infinite interval. On the remaining compact interval, the entire density series has a rational three-step recurrence after three gamma constants are enclosed. We bound the infinite tail and convert a finite lower polynomial into Bernstein form. All coefficient sign decisions are exact rational comparisons. This supplies the positive probability density; failure of log-ID then follows from a separate integer inequality.

2 An explicit signed beta mixture and its positivity

Throughout this section, $\beta > 0$, and we write

$$F_\beta(w) = F^{(2)}_{1/3,\beta}(w) = \sum_{n=0}^{\infty} \frac{w^n}{\Gamma(\beta + n/3)^2}, \qquad \mathcal{F}_\beta(w) = \Gamma(\beta)^2 F_\beta(w).$$

The normalization is $\mathcal{F}_\beta(0) = 1$. The construction below is valid before positivity has been established: its output is initially a finite signed density. In particular, no existence of a random variable having the desired Mellin transform is assumed.

Lemma 2.1 (A signed density from two beta variables)**.** *Let B_1, B_2, G be independent, with*

$$B_1 \sim \mathrm{Beta}\Big(\frac{1}{3}, \beta + \frac{2}{3}\Big), \qquad B_2 \sim \mathrm{Beta}\Big(\frac{2}{3}, \beta + \frac{1}{3}\Big), \qquad G \sim \mathrm{Exp}(1),$$

and put $B = B_1 B_2$. Define

$$P_\beta(y) = y^2 + (2\beta - 3)y + (\beta - 1)^2$$

and, for $z > 0$,

$$h_\beta(z) = \frac{1}{\beta^2}\,\mathbb{E}\Big[B^{-1} e^{-z/B} P_\beta(z/B)\Big]. \tag{2.1}$$

This is a smooth finite signed density of total mass one. Its Mellin integral is absolutely convergent for $\Re s > -1/3$, and equals

$$\begin{aligned}\int_0^\infty z^s h_\beta(z)\,dz &= \frac{\Gamma(\beta)^2\Gamma(s+\frac{1}{3})\Gamma(s+\frac{2}{3})\Gamma(s+1)}{\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})\Gamma(s+\beta)^2} \\ &= 27^{-s}\frac{\Gamma(1+3s)\Gamma(\beta)^2}{\Gamma(\beta+s)^2}.\end{aligned} \tag{2.2}$$

Consequently the change of variables $z = x^3/27$ gives a smooth signed density on $(0, \infty)$,

$$q_\beta(x) = \frac{x^2}{9}\, h_\beta(x^3/27), \tag{2.3}$$

whose absolutely convergent Mellin transform is

$$M_\beta(s) := \int_0^\infty x^s q_\beta(x)\,dx = \frac{\Gamma(1+s)\Gamma(\beta)^2}{\Gamma(\beta + s/3)^2}, \qquad \Re s > -1. \tag{2.4}$$

The reciprocals of Gamma functions in these formulas are understood as entire functions, so zeros of a denominator cause no ambiguity.

Proof. For $\sigma > -1/3$, the beta integrals give

$$\mathbb{E}B^s = \frac{\Gamma(s+\frac{1}{3})\Gamma(s+\frac{2}{3})\Gamma(\beta+1)^2}{\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})\Gamma(s+\beta+1)^2}, \qquad \Re s = \sigma. \tag{2.5}$$

In particular, $\mathbb{E}B^\sigma < \infty$. Substituting $z = By$ and applying Tonelli to absolute values yields

$$\int_0^\infty z^\sigma |h_\beta(z)|\,dz \le \frac{\mathbb{E}B^\sigma}{\beta^2}\int_0^\infty y^\sigma e^{-y}|P_\beta(y)|\,dy < \infty.$$

This justifies Fubini also for complex s in the stated half-plane. The elementary Gamma integral is

$$\begin{aligned}\int_0^\infty y^s e^{-y} P_\beta(y)\,dy &= \Gamma(s+1)\big((s+1)(s+2) + (2\beta-3)(s+1) + (\beta-1)^2\big) \\ &= \Gamma(s+1)(s+\beta)^2.\end{aligned}$$

Combining this with (2.5) and $\Gamma(\beta+1) = \beta\Gamma(\beta)$ proves the first equality in (2.2). Gauss's multiplication formula with multiplier three, [2, §5.5(iii), Eq. 5.5.6], gives the second equality. At $s = 0$ the integral is one. Smoothness on every compact subinterval of $(0, \infty)$ follows by differentiating under the expectation: the exponential suppresses every negative power of B when z stays bounded away from zero.

Finally, $dz/dx = x^2/9$, and

$$\int_0^\infty x^s q_\beta(x)\,dx = 27^{s/3}\int_0^\infty z^{s/3} h_\beta(z)\,dz.$$

The factors $27^{s/3}$ and $27^{-s/3}$ cancel exactly, giving (2.4) and its absolute convergence for $\Re s > -1$. □

The signed-density formula can also be expressed by a differential operator acting on an ordinary positive density. Indeed, $W = BG$ has density

$$f_W(z) = \mathbb{E}[B^{-1}e^{-z/B}].$$

For $L_\beta = -z\,d/dz + \beta - 1$, differentiation under this expectation gives

$$h_\beta = \beta^{-2}L_\beta^2 f_W. \tag{2.6}$$

The preceding proof establishes the Mellin identity directly, so (2.6) does not require any unproved integration-by-parts boundary conditions.

Lemma 2.2 (Entire transform and the positivity criterion)**.** *For every $a > 0$,*

$$\int_0^\infty e^{ax}|q_\beta(x)|\,dx < \infty.$$

Consequently, for every $w \in \mathbb{C}$,

$$\int_0^\infty e^{wx}q_\beta(x)\,dx = \Gamma(\beta)^2\sum_{n=0}^\infty \frac{w^n}{\Gamma(\beta+n/3)^2} = \mathcal{F}_\beta(w). \tag{2.7}$$

In particular,

$$F_\beta(-\lambda) \text{ is completely monotone for } \lambda > 0 \quad\Longleftrightarrow\quad q_\beta(x) \geq 0 \text{ for every } x > 0. \tag{2.8}$$

Proof. For $y \geq 0$, the function $y|P_\beta(y)|e^{-y/2}$ is bounded. Since $y = z/B \geq z$, (2.1) therefore gives, for $z \geq 1$,

$$|h_\beta(z)| \leq C_\beta z^{-1}e^{-z/2}.$$

Thus $|q_\beta(x)| \leq C'_\beta x^{-1}e^{-x^3/54}$ for $x \geq 3$. Near zero, absolute integrability already follows from Lemma 2.1. This proves all the asserted absolute exponential moments. Expanding the exponential and using (2.4) at nonnegative integers now proves (2.7) by absolute convergence.

If $q_\beta \geq 0$, its mass is one and (2.7) is a probability Laplace transform. Conversely, complete monotonicity and $\mathcal{F}_\beta(0) = 1$ give a probability measure on $[0,\infty)$ by Bernstein's theorem. Its Laplace transform equals that of the finite signed measure $q_\beta(x)\,dx$. Uniqueness of finite-measure Laplace transforms identifies the two measures. Continuity of q_β then gives pointwise nonnegativity. This also proves uniqueness of the representing probability law without invoking a moment-indeterminacy argument. □

Lemma 2.3 (The density series, justified by a finite contour shift)**.** *For every $\beta > 0$, the signed density extends to an entire function of x, with the convergent series*

$$q_\beta(x) = \Gamma(\beta)^2 g_\beta(x), \qquad g_\beta(x) = \sum_{n=0}^\infty \frac{(-x)^n}{n!\,\Gamma(\beta-(n+1)/3)^2}. \tag{2.9}$$

It satisfies the regular differential equation

$$9q'''_\beta(x) + x^2q''_\beta(x) + (9-6\beta)xq'_\beta(x) + (4-3\beta)^2q_\beta(x) = 0. \tag{2.10}$$

Proof. Stirling's estimate on vertical strips, [2, §5.11], gives

$$M_\beta(c+it) = O\Big((1+|t|)^{c/3+3/2-2\beta}e^{-\pi|t|/6}\Big), \qquad |t| \longrightarrow \infty,$$

uniformly for c in any fixed bounded real interval. Mellin inversion consequently holds absolutely on the line $c = 0$:

$$q_\beta(x) = \frac{1}{2\pi i}\int_{-i\infty}^{i\infty} M_\beta(s)x^{-s-1}\,ds.$$

Move this contour only as far as $\Re s = -9/2$. The horizontal integrals tend to zero by the displayed estimate. The only possible poles crossed are $-1, -2, -3, -4$, and their residues give

$$q_\beta(x) = \Gamma(\beta)^2 \sum_{n=0}^{3} \frac{(-x)^n}{n!\,\Gamma(\beta-(n+1)/3)^2} + R_\beta(x).$$

A reciprocal-Gamma zero can cancel a pole, in which case the same residue formula simply gives zero. Differentiating the remaining absolutely convergent integral shows

$$R_\beta^{(k)}(x) = O(x^{7/2-k}), \qquad 0 \le k \le 3, \quad x \downarrow 0.$$

Hence q_β has a C^3 extension to zero and, for $k = 0, 1, 2$,

$$q_\beta^{(k)}(0) = \frac{\Gamma(\beta)^2(-1)^k}{\Gamma(\beta-(k+1)/3)^2}. \tag{2.11}$$

Gamma's recurrence gives the meromorphic identity

$$(s+3\beta-3)^2 M_\beta(s) = 9s(s-1)(s-2)M_\beta(s-3).$$

For $\Re s > 2$, the Mellin transform of the left side of (2.10), evaluated by integration by parts, is therefore

$$-9s(s-1)(s-2)M_\beta(s-3) + (s+3\beta-3)^2 M_\beta(s) = 0.$$

All the boundary terms vanish. At zero this follows from the C^3 extension, and at infinity it follows either from the beta mixture or from differentiated Mellin inversion on arbitrarily large positive abscissae. Mellin-transform uniqueness proves (2.10) on $(0, \infty)$, and continuity extends it to zero.

The coefficients

$$a_n = \frac{\Gamma(\beta)^2(-1)^n}{n!\,\Gamma(\beta-(n+1)/3)^2}$$

satisfy, including when a coefficient vanishes,

$$a_{n+3} = -\frac{(n+4-3\beta)^2}{9(n+1)(n+2)(n+3)}\, a_n. \tag{2.12}$$

Each of the three residue-class series is entire by the ratio test. Their sum satisfies (2.10) and the three initial conditions (2.11). The leading coefficient of (2.10) is the nonzero constant nine. Uniqueness for this ordinary initial-value problem identifies its entire series solution with q_β for all $x \ge 0$. Thus the proof uses one finite contour shift and does not close a Mellin contour through infinitely many poles. □

Corollary 2.4 (Every nonnegative contact is quadratic)**.** *If $q_\beta \ge 0$ on $(0, \infty)$ and $q_\beta(x_0) = 0$ at $x_0 > 0$, then*

$$q_\beta'(x_0) = 0, \qquad q_\beta''(x_0) > 0.$$

Proof. Nonnegativity gives $q_\beta'(x_0) = 0$ and $q_\beta''(x_0) \ge 0$. If the second derivative were zero, uniqueness for the regular third-order equation (2.10) would give $q_\beta \equiv 0$, contrary to its total mass one. □

Lemma 2.5 (An elementary positive tail)**.** *For every $\beta \ge \sqrt{2} - 1$,*

$$q_\beta(x) > 0 \qquad \text{whenever } x \ge \sqrt[3]{54}.$$

In particular this holds for every $\beta \ge 5/12$. At the principal rational parameter $\beta = 3/7$,

$$P_{3/7}(y) = y^2 - \frac{15}{7}y + \frac{16}{49}, \qquad P_{3/7}(2) = \frac{2}{49} > 0.$$

Proof. For $y \geq 2$,

$$P_\beta(2) = \beta^2 + 2\beta - 1 \geq 0, \qquad P'_\beta(y) = 2y + 2\beta - 3 \geq 1 + 2\beta > 0.$$

Since $0 < B < 1$ almost surely, $z \geq 2$ implies $z/B > 2$. The integrand in (2.1) is then strictly positive. Equation (2.3) completes the proof. The rational sufficient bound follows also from $P_{5/12}(2) = 1/144 > 0$. □

Lemma 2.6 (Transfer to all larger beta parameters)**.** *Suppose $q_\beta \geq 0$, and let $\beta' > \beta$. Take independent*

$$U_1, U_2 \sim \mathrm{Beta}(\beta, \beta' - \beta), \qquad R = (U_1 U_2)^{1/3},$$

independent also of a random variable X_β with density q_β. Then

$$X_{\beta'} \stackrel{d}{=} X_\beta R \tag{2.13}$$

has density $q_{\beta'}$. In particular $F_{\beta'}(-\lambda)$ is completely monotone. If q_β is strictly positive on a terminal interval of the positive half-line, then $q_{\beta'}$ is strictly positive at every positive point.

Proof. The ordinary beta integral gives

$$\mathbb{E}R^s = \left(\frac{\Gamma(\beta')\Gamma(\beta + s/3)}{\Gamma(\beta)\Gamma(\beta' + s/3)}\right)^2, \qquad \Re s > -3\beta.$$

Multiplication by $M_\beta(s)$ yields $M_{\beta'}(s)$. Taking $s = it$, uniqueness of Fourier transforms after the change of variables $x = e^y$ identifies the product law with the finite signed measure $q_{\beta'}(x)\,dx$; hence the latter is positive. Equivalently, its continuous density is

$$q_{\beta'}(x) = \mathbb{E}[R^{-1} q_\beta(x/R)].$$

The law of R has support $(0, 1)$. For every fixed $x > 0$, the event that x/R lies in the positive terminal interval has positive probability. The last formula is therefore strictly positive. Complete monotonicity follows from Lemma 2.2. □

Lemma 2.7 (An exact obstruction to logarithmic infinite divisibility)**.** *If $0 < \beta \leq 13/30$ and $q_\beta \geq 0$, a random variable X_β with density q_β is not logarithmically infinitely divisible.*

Proof. The Gamma logarithm formula used in [1, Section 2] gives

$$\log M_\beta(it) = ic_\beta t + \int_0^\infty (e^{-itr} - 1 + itr)\frac{\phi_\beta(r)}{r}\,dr,$$

where $c_\beta = \psi(1) - \frac{2}{3}\psi(\beta)$ and

$$\phi_\beta(r) = \frac{e^{-r}}{1 - e^{-r}} - 2\frac{e^{-3\beta r}}{1 - e^{-3r}}. \tag{2.14}$$

The integral is absolutely convergent after compensation, and $\int_0^\infty r|\phi_\beta(r)|\,dr < \infty$. If $\log X_\beta$ were infinitely divisible, its Lévy measure would have to be nonnegative, forcing $\phi_\beta \geq 0$. For completeness, this necessity follows directly from uniqueness: the negative second derivative of the displayed exponent is the Fourier transform of the finite signed measure supported at $-r$ with mass element $r\phi_\beta(r)\,dr$. On the other hand, the negative second derivative of an infinitely divisible exponent with finite variance is the Fourier transform of its nonnegative variance measure, including any Gaussian mass at zero. Fourier uniqueness thus forces $r\phi_\beta(r)\,dr$ to be nonnegative. The required finite variance of $\log X_\beta$ follows from (2.4) at small positive and negative real arguments.

Put $u = e^{-r}$. Nonnegativity in (2.14) is equivalent to

$$2u^{3\beta-1} \le 1 + u + u^2, \qquad 0 < u < 1. \tag{2.15}$$

At $u = 1/4$ and $\beta \le 13/30$, the left side is at least $2^{2/5}$, whereas the right side is $21/16$. The strict reverse inequality is certified by the exact integer comparison

$$2^{22} = 4\,194\,304 > 4\,084\,101 = 21^5.$$

Consequently $\phi_\beta(\log 4) < 0$, proving the assertion. At $\beta = 3/7$ one can alternatively use the particularly simple comparison

$$21^7 = 1\,801\,088\,541 < 2\,147\,483\,648 = 2^{31}.$$

□

Corollary 2.8 (An explicit interval of counterexamples)**.** *For every*

$$\frac{3}{7} \le \beta \le \frac{13}{30},$$

the function $F^{(2)}_{1/3,\beta}(-\lambda)$ is completely monotone on $(0,\infty)$, and its normalized representing law has a strictly positive density and is not logarithmically infinitely divisible. Complete monotonicity in fact holds for every $\beta \ge 3/7$.

Proof. The finite certificate accompanying this proof establishes

$$g_{3/7}(x) > \frac{1}{20000} \qquad (0 \le x \le 4). \tag{2.16}$$

It uses the degree-89 truncation of (2.9), a rigorous three-step recurrence remainder, and four rational Bernstein subintervals. The certificate and its independent verification are supplied in the computational section; the analytical identity needed by that certificate is proved in Lemma 2.3. Since $\sqrt[3]{54} < 4$, (2.16) and Lemma 2.5 cover the whole positive half-line. Lemmas 2.2 and 2.6 prove complete monotonicity and strict density positivity for every larger parameter. Lemma 2.7 gives the stated interval on which logarithmic infinite divisibility fails. □

3 Finite certificates for an infinite interval problem

This section supplies the compact-interval input to the analytical construction. It also proves the lower boundary obstruction and the uniform estimates used to isolate its contact point. The computations are proofs by interval enclosure and exact polynomial inequalities, rather than tests of sampled approximate values.

3.1 Three gamma constants and a rational recurrence

Write $g_\beta(x) = \sum_{n\ge0}(-1)^n a_n(\beta)x^n$, where

$$a_n(\beta) = \frac{1}{n!\Gamma(\beta - (n+1)/3)^2} \ge 0.$$

The gamma recurrence gives, exactly,

$$a_{n+3}(\beta) = a_n(\beta)\frac{(n+4-3\beta)^2}{9(n+1)(n+2)(n+3)}. \tag{3.1}$$

All coefficients are therefore determined by a_0, a_1, a_2. For $\beta \in [\ell, u] \subset (1/3, 1/2)$, the positive multiplier in (3.1) decreases as β increases. If $A_n^- \le a_n(\beta) \le A_n^+$ throughout that parameter interval, valid next enclosures are

$$A_{n+3}^- = A_n^- \frac{(n+4-3u)^2}{9(n+1)(n+2)(n+3)}, \quad A_{n+3}^+ = A_n^+ \frac{(n+4-3\ell)^2}{9(n+1)(n+2)(n+3)}.$$

The program first encloses the three reciprocal-gamma squares with 384-bit Arb arithmetic [3], then explicitly verifies strict rational lower and upper bounds with denominator 10^{80}. All subsequent operations use exact rational numbers.

3.2 A uniform remainder bound, including derivatives

Take $N = 90$. For $d \in \{0, 1, 2\}$ define $T_n^{(d)} = a_n(\beta) n^{\underline{d}} 4^{n-d}$, with $n^{\underline{d}} = n!/(n-d)!$. For $n \ge 90$,

$$\begin{aligned} \frac{T_{n+3}^{(d)}}{T_n^{(d)}} &= \frac{64(n+4-3\beta)^2}{9(n+1-d)(n+2-d)(n+3-d)} \\ &\le \frac{64(n+3)^2}{9(n-1)n(n+1)} \le \frac{64 \cdot 93^2}{9 \cdot 89 \cdot 90 \cdot 91} < \frac{1}{10}. \end{aligned} \tag{3.2}$$

The rational function in the middle is decreasing for $n \ge 90$; this follows, for example, from its logarithmic derivative $2/(n+3) - 1/(n-1) - 1/n - 1/(n+1) < 0$. Summing the three residue classes modulo three gives

$$\sup_{0 \le x \le 4} \left| g_\beta^{(d)}(x) - \sum_{n=d}^{89} (-1)^n a_n(\beta) n^{\underline{d}} x^{n-d} \right| \le E_d := \frac{10}{9} \sum_{n=90}^{92} A_n^+ n^{\underline{d}} 4^{n-d}. \tag{3.3}$$

Every displayed application satisfies $E_0 < 10^{-20}$; the parameter-box second derivative satisfies $E_2 < 10^{-16}$. These comparisons are themselves checked with exact rational arithmetic.

A rational lower polynomial for $g_\beta^{(d)}$ is obtained by using A_n^- for even n, A_n^+ for odd n, and subtracting E_d from the constant coefficient. An upper polynomial uses the opposite endpoints and adds E_d. Positivity of x is the only order property needed for these substitutions.

3.3 Bernstein positivity and a verifiable cover

Let P be one of the rational lower polynomials, of degree at most m, and let $[a, b]$ be a rational interval. After substitution,

$$P(a + (b-a)t) = \sum_{k=0}^{m} v_k t^k = \sum_{j=0}^{m} B_j \binom{m}{j} t^j (1-t)^{m-j}, \quad B_j = \sum_{k=0}^{j} v_k \frac{\binom{j}{k}}{\binom{m}{k}}.$$

The Bernstein basis functions are nonnegative and sum to one on $[0, 1]$. Hence $B_j > c$ for every j proves $P > c$ throughout $[a, b]$. If this test fails, de Casteljau subdivision at the midpoint gives the exact Bernstein coefficients on the two halves. The verifier checks that the final closed intervals cover the whole requested interval without gaps.

Proposition 3.1 (Certified compact bounds)**.** *The following strict inequalities hold:*

$$g_{3/7}(x) > \frac{1}{20000} \qquad (0 \le x \le 4), \tag{3.4}$$

$$g_{b_{\mathrm{U}}}(x) > 10^{-13} \qquad (0 \le x \le 4), \tag{3.5}$$

$$g_{b_{\mathrm{L}}}\left(\frac{511574841194}{10^{12}}\right) < -10^{-13}, \tag{3.6}$$

$$g_\beta(x) > 10^{-6} \qquad (\beta \in [b_{\mathrm{L}}, b_{\mathrm{U}}], \quad x \in [0, 1/2] \cup [53/100, 4]), \tag{3.7}$$

$$g_\beta''(x) > \frac{1}{20} \qquad (\beta \in [b_{\mathrm{L}}, b_{\mathrm{U}}], \quad x \in [1/2, 53/100]). \tag{3.8}$$

Proof. Apply the preceding rational enclosures, (3.3), and Bernstein coefficient test. The complete executable verifier appears in Appendix A; the exact inputs are the displayed rational numbers. Its successful output is summarized in Table 1. For (3.6), evaluate the upper polynomial at the stated rational point. There is no floating-point sign decision in this step. □

Parameter or box	Interval	Certified lower bound	Pieces
$\beta = 3/7$	$[0, 4]$	$g > 1/20000$	4
$\beta = b_{\mathrm{U}}$	$[0, 4]$	$g > 10^{-13}$	18
$\beta \in [b_{\mathrm{L}}, b_{\mathrm{U}}]$	$[0, 1/2]$	$g > 10^{-6}$	1
$\beta \in [b_{\mathrm{L}}, b_{\mathrm{U}}]$	$[53/100, 4]$	$g > 10^{-6}$	1
$\beta \in [b_{\mathrm{L}}, b_{\mathrm{U}}]$	$[1/2, 53/100]$	$g'' > 1/20$	1

Table 1: Exact rational Bernstein certificates after gamma enclosures. The degree is 89 for g, and 87 for g''.

For the small rational counterexample, the four pieces are simply $[0, 1/2]$, $[1/2, 1]$, $[1, 2]$, and $[2, 4]$. For $\beta = b_{\mathrm{U}}$, the partition endpoints, in increasing order, are

$$0,\ \tfrac{1}{2},\ \tfrac{65}{128},\ \tfrac{261}{512},\ \tfrac{523}{1024},\ \tfrac{1047}{2048},\ \tfrac{2095}{4096},\ \tfrac{8381}{16384},\ \tfrac{16763}{32768},\ \tfrac{4191}{8192},\ \tfrac{131}{256},\ \tfrac{33}{64},\ \tfrac{17}{32},\ \tfrac{9}{16},\ \tfrac{5}{8},\ \tfrac{3}{4},\ 1,\ 2,\ 4.$$

Thus the near-zero minimum is covered by a finite interval proof; the argument does not infer global positivity from the decimal value of a numerical minimizer.

3.4 An independent compact-interval check

For $\beta = 3/7$, the same conclusion was checked independently by a grid with a rigorous derivative bound and a rigorous infinite-tail bound. On a unit interval subdivided into K parts, assume every grid point has $g > L$ and the whole interval has $|g'| < M$. Every real point is within $1/(2K)$ of a grid point, so the mean value theorem gives $g > L - M/(2K)$. A rigorous derivative majorant on $[0, X]$, for $X \in \{1, 2, 3, 4\}$, is

$$\sum_{n=1}^{89} n a_n X^{n-1} + \frac{1}{X} \sum_{n=90}^{92} a_n X^n \left(\frac{n}{1-R} + \frac{3R}{(1-R)^2} \right), \qquad R = \frac{2}{25}.$$

Indeed, the undifferentiated three-step ratio is at most $64 \cdot 93/(9 \cdot 91 \cdot 92) < R$, and $\sum_{k \geq 0}(n + 3k)R^k = n/(1-R) + 3R/(1-R)^2$. The finite sums and gamma constants are enclosed using Arb. The verified values are

Interval	K	L	M	$L - M/(2K)$
$[0, 1]$	2048	9/100000	1/7	4939/89600000
$[1, 2]$	64	9/1000	1/2	163/32000
$[2, 3]$	64	3/50	3	117/3200
$[3, 4]$	512	1/20	29	111/5120

All four rational lower bounds exceed 1/20000. There are 2692 grid evaluations, including repeated endpoints. This independent method is included in Appendix B. The Bernstein certificate already suffices for the theorem.

4 The boundary and its unique quadratic contact

We now prove Theorem 1.2 without assuming the qualitative strict-gap theorem from the corpus. The signed-density construction and Laplace-transform uniqueness give

$$F^{(2)}_{1/3,\beta}(-x) \text{ is CM} \quad \Longleftrightarrow \quad q_\beta(x) \geq 0 \text{ for every } x > 0. \tag{4.1}$$

Indeed, a nonnegative signed density of mass one is a probability density. Conversely, the Bernstein representing positive measure and $q_\beta(x)\,\mathrm{d}x$ would have the same Laplace transform and hence are equal. If q_β is negative at a point, continuity excludes such a positive measure.

The function $(\beta, x) \mapsto q_\beta(x)$ is jointly continuous on compact subsets of $(0,\infty)\times[0,\infty)$. Indeed, the three initial reciprocal-gamma coefficients are continuous in β, and the recurrence has ratio $O(1/n)$ uniformly on bounded parameter intervals. The three residue-class series thus converge uniformly on compact sets. This supplies the compact continuity used below, including at $x = 0$.

The beta-factor transfer proved above makes the CM set an upper set in $\beta > 0$. It is nonempty by (3.5) and the positive tail; it does not contain b_{L} by (3.6). Let β_* be its infimum. It follows that $\beta_* \in [b_{\mathrm{L}}, b_{\mathrm{U}}]$, and pointwise continuity of q_β in β shows $q_{\beta_*} \geq 0$. Thus β_* itself is in the CM set.

The lower inequality is strict because $q_{b_{\mathrm{L}}}$ is negative at the certified point. For the upper inequality, $q_{b_{\mathrm{U}}}$ has a positive minimum on $[0,4]$ by (3.5). Compact continuity preserves positivity when β decreases slightly. The tail remains positive uniformly for all these parameters: they can be kept above $5/12$, and

$$(5/12)^2 + 2(5/12) - 1 = 1/144 > 0.$$

The beta-mixture tail criterion therefore applies on $x^3 \geq 54$, which overlaps $[0,4]$. A parameter strictly below b_{U} is CM, proving $\beta_* < b_{\mathrm{U}}$.

If q_{β_*} were positive throughout $[0,4]$, the same compact-continuity and positive-tail argument would yield a CM parameter below β_*, a contradiction. Hence it has a zero. The uniform bound (3.7) and the positive tail place every zero in $(1/2, 53/100)$. On that interval, (3.8) makes g_{β_*} strictly convex. A nonnegative strictly convex function has at most one zero. At the zero, its derivative vanishes and its second derivative exceeds $1/20$, proving the asserted order and inequality for q''.

To prove uniqueness of the pair solving $g = \partial_x g = 0$ in the displayed rectangle, suppose such a pair (β, x) exists. Strict convexity makes x the unique minimum on the contact interval; the outside and tail bounds then give $q_\beta \geq 0$ everywhere. Thus $\beta \geq \beta_*$. If $\beta > \beta_*$, beta-factor transfer from β_* makes the new density strictly positive: the scaling factor has a density positive on $(0,1)$, and the original density vanishes at only one point. This contradicts $g_\beta(x) = 0$. Consequently $\beta = \beta_*$ and $x = x_*$.

Finally, $\beta \geq b_{\mathrm{U}}$ implies CM by beta-factor transfer, whereas $\beta \leq 13/30$ violates the scalar log-ID condition at $z = 1/64$. This proves the explicit interval of counterexamples.

4.1 The birth of the unique negative interval

Theorem 4.1 (A transverse quadratic contact)**.** *Let (β_*, x_*) be the unique critical pair certified above, and put*

$$a = \frac{1}{2}\partial_x^2 q_{\beta_*}(x_*) > 0, \qquad b = \partial_\beta q_\beta(x_*)|_{\beta=\beta_*}.$$

Then $b > 0$. For every sufficiently small $\varepsilon > 0$, the density $q_{\beta_*-\varepsilon}$ is negative on exactly one interval $(x_-(\varepsilon), x_+(\varepsilon))$, where

$$x_\pm(\varepsilon) = x_* \pm \sqrt{b/a}\,\varepsilon^{1/2} + O(\varepsilon). \tag{4.2}$$

Its total negative mass satisfies

$$\int_0^\infty \max\{-q_{\beta_*-\varepsilon}(x), 0\}\,dx = \frac{4b^{3/2}}{3\sqrt{a}}\,\varepsilon^{3/2} + o(\varepsilon^{3/2}). \tag{4.3}$$

Proof. Differentiating (2.4), using the digamma integral, and applying Mellin inversion gives

$$\partial_\beta q_\beta(x) = 2\int_0^\infty \frac{e^{-\beta t}}{1-e^{-t}}\left[e^{t/3} q_\beta(x e^{t/3}) - q_\beta(x)\right] dt. \tag{4.4}$$

Here β may range over a compact neighborhood of β_*. To justify the calculation, on $s = i\tau$ the identity being inverted is

$$\partial_\beta M_\beta(s) = 2M_\beta(s) \int_0^\infty \frac{e^{-\beta t}(e^{-st/3} - 1)}{1 - e^{-t}}\, dt.$$

The bound $|e^{-i\tau t/3} - 1| \le \min\{2, |\tau|t/3\}$, together with the uniform exponential decay of $M_\beta(i\tau)$, justifies Fubini and differentiation under inversion. The resulting real-space integral is also absolutely convergent: its bracket is $O(t)$ at zero, while the beta-mixture tail estimate and $e^{-\beta t}$ control infinity. At the critical zero, the subtracted term vanishes and $q_{\beta_*}(xe^{t/3}) > 0$ for $x = x_*$ and every $t > 0$. Thus (4.4) proves $b > 0$.

The density is jointly real analytic in (β, x) near (β_*, x_*), as follows from (2.9) and its locally uniformly convergent coefficient recurrence. Hence

$$q_{\beta_* - \varepsilon}(x_* + h) = ah^2 - b\varepsilon + O(|h|^3 + \varepsilon|h| + \varepsilon^2).$$

Set $\delta = \sqrt{\varepsilon}$ and $h = \delta y$. The quotient $\delta^{-2} q_{\beta_* - \delta^2}(x_* + \delta y)$ extends analytically to $\delta = 0$, with value $ay^2 - b$. The implicit function theorem at its two simple zeros $y = \pm\sqrt{b/a}$ proves (4.2). The certified positivity outside $(1/2, 53/100)$, together with the certified strict convexity inside this interval for all parameters in the enclosing beta box, shows that these are the only two zeros and that their intervening interval is the whole negative set. Finally, the same rescaling and the Taylor estimate give

$$\varepsilon^{-3/2} \int_0^\infty \max\{-q_{\beta_* - \varepsilon}(x), 0\}\, dx \longrightarrow \int_{-\sqrt{b/a}}^{\sqrt{b/a}} (b - ay^2)\, dy = \frac{4b^{3/2}}{3\sqrt{a}},$$

which proves (4.3). □

5 Scope of the conclusion

The literal universal equivalence in (1.1) is false; one rational triple already suffices to establish that statement. The interval of examples is a strengthening of the same resolution, not a list of separate conjectures. The full positive-parameter classification of $F^{(\gamma)}_{\alpha,\beta}$ is not supplied by this result. At $(\alpha, \gamma) = (1/3, 2)$, however, the boundary is uniquely specified by the double-contact equations in a rigorously isolated rectangle, with a parameter enclosure of width 10^{-12}.

The method replaces an unbounded positivity check by two complementary finite objects: a quadratic polynomial that controls the whole tail, and a rational Bernstein certificate on a compact interval. Its mathematical dependency on special-function numerics is restricted to rigorous enclosures of three gamma constants for each fixed parameter or parameter interval. The supplied source reproduces the certificate, including the uniform boundary estimates. No external peer-review or formal proof-assistant validation is asserted.

References

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[2] NIST Digital Library of Mathematical Functions, Chapter 5, §5.5(iii), Eq. 5.5.6 (multiplication formula), and §5.11 (asymptotic expansions of the gamma function). https://dlmf.nist.gov/5.5; https://dlmf.nist.gov/5.11.

[3] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Transactions on Computers **66**(8) (2017), 1281–1292. doi:10.1109/TC.2017.2690633. https://arxiv.org/abs/1611.02831.

[4] Keio Research Notes, *Complete monotonicity beyond logarithmic infinite divisibility: a counterexample theorem for Le Roy functions*, 3 November 2026. Unrefereed research manuscript [116] in the Verifiable Mathematical Reasoning Corpus, https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus, snapshot c30d40119134a59eda6fa5cc6331f27c8e67d034.

[5] Keio Research Notes, *Equivalence Bridges from the 130-Manuscript Corpus: explicit independence, arithmetic obstructions, and a positivity boundary*, 8 November 2026. Preceding research synthesis, 34 pages; the Le Roy section supplies the earlier one-sided bound $\beta_c(1/3, 2) > 0.428292245841$.

A Complete exact-rational certificate

The single `.tex` source contains both verification programs in `filecontents*` environments. Compiling it writes the programs alongside the compilation output. They require Python 3 and `python-flint`; the recorded runs used `python-flint 0.9.0`. Run

```
python3 leroy_explicit_certificate.py
python3 leroy_independent_certificate.py
```

The first program checks every inequality in Proposition 3.1 and writes a JSON record containing the exact rational gamma enclosures, tail bounds, and covering intervals. It uses Arb only to obtain and verify the initial rational enclosures. All polynomial bounds and subdivisions use exact rational arithmetic. The successful run and an independently executed copy produced identical JSON records. The audited primary program has SHA256

`2af3eca6a5979ef1df0188b32f1425a2a0a0813d9ec407ed80c111cf370e2d60.`

The following is the complete primary program.

```
"""Exact Bernstein certificate for explicit Le Roy counterexamples.

Only three reciprocal-gamma evaluations per parameter enclosure use Arb.
Everything after the checked rational enclosures is exact fmpq arithmetic.
Requires python-flint. No floating-point value is used to decide a sign.
"""
import json
import math
from pathlib import Path
import flint
from flint import arb, fmpq as Q, ctx

ctx.prec = 384
N = 90
DIGITS = 80
DEN = 10 ** DIGITS
BL = Q(428292245841, 10**12)
BU = Q(428292245842, 10**12)
records = []


def rational_enclosure(v):
    a = (v.lower() * DEN).floor().unique_fmpz()
    b = (v.upper() * DEN).floor().unique_fmpz()
    assert a is not None and b is not None
    lo, hi = Q(int(a)-1, DEN), Q(int(b)+2, DEN)
    assert arb(lo) < v and v < arb(hi)
    return lo, hi


def coefficients(bl, bu):
    bb = arb(bl).union(arb(bu))
    lo, hi = [], []
    seeds = []
    for r in range(3):
        v = (bb - arb(Q(r+1, 3))).rgamma()**2 / math.factorial(r)
        l, u = rational_enclosure(v)
        assert l > 0
        lo.append(l)
        hi.append(u)
        seeds.append([str(l), str(u)])
    for n in range(N):
        den = 9*(n+1)*(n+2)*(n+3)
        lo.append(lo[n]*(n+4-3*bu)**2/den)
        hi.append(hi[n]*(n+4-3*bl)**2/den)
    return lo, hi, seeds


def falling(n, d):
    return math.factorial(n)//math.factorial(n-d)
```

```
def tail_bound(hi, d):
    # For n>=90 and 1/3<beta<1/2, the three-step term ratio,
    # even after d<=2 derivatives, is at most the expression below.
    assert 0 <= d <= 2
    r = Q(64*(N+3)**2, 9*(N+1)*N*(N-1))
    assert r < Q(1, 10)
    return sum(hi[n]*falling(n, d)*4**(n-d)
               for n in range(N, N+3))*Q(10, 9)


def lower_polynomial(lo, hi, d=0):
    p = [((-1)**n)*(lo[n] if n % 2 == 0 else hi[n])*falling(n, d)
         for n in range(d, N)]
    e = tail_bound(hi, d)
    p[0] -= e
    return p, e


def bernstein_on_interval(p, a, b):
    # p(a+(b-a)t)=sum a_k t^k=sum b_j binom(m,j)t^j(1-t)^(m-j).
    m = len(p)-1
    aa = [a**i for i in range(m+1)]
    width = b-a
    power = [sum(p[k]*math.comb(k,j)*aa[k-j]
                 for k in range(j,m+1))*width**j for j in range(m+1)]
    return [sum(power[k]*Q(math.comb(j,k),math.comb(m,k))
                for k in range(j+1)) for j in range(m+1)]


def split_bernstein(b):
    row = b[:]
    left, right = [row[0]], [row[-1]]
    while len(row)>1:
        row = [(row[i]+row[i+1])/2 for i in range(len(row)-1)]
        left.append(row[0])
        right.append(row[-1])
    return left, list(reversed(right))


def certify_positive(p, a, b, target, name):
    initial = bernstein_on_interval(p, a, b)
    stack = [(a,b,initial,0)]
    leaves = []
    minimum = None
    while stack:
        a0,b0,bb,depth = stack.pop()
        lb = min(bb)
        if lb > target:
            leaves.append([str(a0),str(b0),depth])
            minimum = lb if minimum is None else min(minimum,lb)
            continue
        assert depth < 32, (name, str(a0), str(b0), str(lb))
        left,right = split_bernstein(bb)
        mid = (a0+b0)/2
        stack.append((mid,b0,right,depth+1))
        stack.append((a0,mid,left,depth+1))
    # Explicit check that the certified closed intervals cover [a,b].
    assert Q(leaves[0][0]) == a and Q(leaves[-1][1]) == b
    assert all(leaves[i][1] == leaves[i+1][0] for i in range(len(leaves)-1))
    print(name, ':', len(leaves), 'leaves; max depth', max(t[2] for t in leaves),
          '; exact lower bound exceeds', target, flush=True)
    records.append({'name':name,'interval':[str(a),str(b)],'target':str(target),
                    'leaves':leaves,'minimum_coefficient':str(minimum)})


def eval_poly(p, x):
    ans = Q(0)
    for a in reversed(p):
        ans = ans*x+a
    return ans
```

```
def main():
    print('python-flint',flint.__version__,'; Arb precision',ctx.prec,flush=True)
    cases = [
        ('beta_3_over_7',Q(3,7),Q(3,7),Q(1,20000)),
        ('beta_upper',BU,BU,Q(1,10**13)),
    ]
    seed_records=[]
    for name,bl,bu,target in cases:
        lo,hi,seeds=coefficients(bl,bu)
        p,e=lower_polynomial(lo,hi)
        assert e < Q(1,10**20)
        certify_positive(p,Q(0),Q(4),target,name)
        assert bl*bl+2*bl-1>0  # positive beta-mixture tail for z>=2
        seed_records.append({'name':name,'beta':[str(bl),str(bu)],'seeds':seeds,
                             'tail_bound':str(e)})

    lo,hi,seeds=coefficients(BL,BL)
    # Upper polynomial, with an absolute uniform remainder bound.
    p=[((-1)**n)*(hi[n] if n%2==0 else lo[n]) for n in range(N)]
    p[0]+=tail_bound(hi,0)
    x=Q(511574841194,10**12)
    upper=eval_poly(p,x)
    assert upper < -Q(1,10**13)
    print('beta_lower: g_beta(x) < -1/10^13 at x =',x,flush=True)
    seed_records.append({'name':'beta_lower','beta':[str(BL),str(BL)],'seeds':seeds,
                         'negative_x':str(x),'upper_bound':str(upper)})

    # Uniform parameter-box certificates isolate the only possible contact.
    lo,hi,seeds=coefficients(BL,BU)
    p,e=lower_polynomial(lo,hi)
    certify_positive(p,Q(0),Q(1,2),Q(1,10**6),'parameter_box_left')
    certify_positive(p,Q(53,100),Q(4),Q(1,10**6),'parameter_box_right')
    p2,e2=lower_polynomial(lo,hi,2)
    assert e2 < Q(1,10**16)
    certify_positive(p2,Q(1,2),Q(53,100),Q(1,20),'parameter_box_second_derivative')
    seed_records.append({'name':'parameter_box','beta':[str(BL),str(BU)],'seeds':seeds,
                         'tail_bound':str(e),'second_derivative_tail':str(e2)})

    assert 21**5 < 2**22
    assert 21**7 < 2**31
    assert BU < Q(3,7) < Q(13,30)
    print('ALL CERTIFICATES PASSED',flush=True)
    result={'precision_bits':ctx.prec,'N':N,'gamma_decimal_digits':DIGITS,
            'parameters':seed_records,'certificates':records}
    out=Path(__file__).with_suffix('.json')
    out.write_text(json.dumps(result,indent=2)+'\n')
    print('Certificate record:',out,flush=True)


if __name__=='__main__':
    main()
```

Recorded successful output

```
python-flint 0.9.0 ; Arb precision 384
beta_3_over_7 : 4 leaves; max depth 3 ; exact lower bound exceeds 1/20000
beta_upper : 18 leaves; max depth 17 ; exact lower bound exceeds 1/10000000000000
beta_lower: g_beta(x) < -1/10^13 at x = 255787420597/500000000000
parameter_box_left : 1 leaves; max depth 0 ; exact lower bound exceeds 1/1000000
parameter_box_right : 1 leaves; max depth 0 ; exact lower bound exceeds 1/1000000
parameter_box_second_derivative : 1 leaves; max depth 0 ; exact lower bound exceeds 1/20
ALL CERTIFICATES PASSED
```

B Independent derivative-and-grid certificate

This second program verifies only the compact positivity assertion for $\beta = 3/7$, using a different method. Every sign comparison and error estimate is a rigorous Arb interval comparison with exact rational inputs. Floating-point conversion appears only when selecting a diagnostic grid point to print; it has no role in a proof decision. The interval-cover argument and all thresholds are proved in Section 3.

```
"""Independent compact positivity certificate for beta=3/7.

The certified raw density is
 g(x) = sum_{n>=0} (-x)^n/(n! Gamma(3/7-(n+1)/3)^2).
The normalized inverse Mellin density is Gamma(3/7)^2*g(x).

Requires python-flint. All interval inputs and comparison thresholds are
exact rational numbers. The proof of the tail estimate and the interval
coverage argument is in the accompanying manuscript.
"""
from flint import arb, arb_poly, fmpq, ctx
import flint


def main():
    ctx.prec = 256
    beta = fmpq(3, 7)
    N = 90
    # Three initial Gamma values, followed by exact rational recurrences.
    coeff = [arb(0) for _ in range(N + 3)]
    coeff[0] = arb(beta - fmpq(1, 3)).rgamma() ** 2
    coeff[1] = -(arb(beta - fmpq(2, 3)).rgamma() ** 2)
    coeff[2] = arb(beta - 1).rgamma() ** 2 / 2
    for n in range(N):
        coeff[n + 3] = -coeff[n] * arb(
            (n + 4 - 3 * beta) ** 2 / (9 * (n + 1) * (n + 2) * (n + 3))
        )
    P = arb_poly(coeff[:N])
    R = fmpq(2, 25)
    # At x <= 4 and n >= N, the positive absolute term ratio is <= R.
    assert fmpq(64 * (N + 3), 9 * (N + 1) * (N + 2)) < R
    global_target = fmpq(1, 20000)
    records = [
        (0, 2048, fmpq(9, 100000), fmpq(1, 7)),
        (1, 64, fmpq(9, 1000), fmpq(1, 2)),
        (2, 64, fmpq(3, 50), fmpq(3)),
        (3, 512, fmpq(1, 20), fmpq(29)),
    ]
    print('python-flint:', flint.__version__)
    print('precision:', ctx.prec, 'bits; truncation:', N, 'terms')
    for left, pieces, grid_lower, derivative_upper in records:
        X = left + 1
        seeds = [abs(coeff[n]) * X ** n for n in range(N, N + 3)]
        tail = sum(seeds, arb(0)) / arb(1 - R)
        # For n=N,N+1,N+2 and k>=0, (n+3k)t_(n+3k)
        # <= (n+3k)t_n R^k. Summing gives the derivative tail below.
        derivative_tail = sum((
            seeds[j] * arb(fmpq(N + j, 1) / (1 - R) + 3 * R / (1 - R) ** 2) / X
            for j in range(3)
        ), arb(0))
        derivative_bound = sum((
            n * abs(coeff[n]) * X ** (n - 1) for n in range(1, N)
        ), arb(0)) + derivative_tail
        assert derivative_bound < arb(derivative_upper)
        minimum = None
        minimizer = None
        for j in range(pieces + 1):
            x = fmpq(left, 1) + fmpq(j, pieces)
            lower = P(arb(x)) - tail
            assert lower > arb(grid_lower), (left, x, lower)
            if minimum is None or float(lower.mid()) < float(minimum.mid()):
                minimum, minimizer = lower, x
        # Every x in [left,left+1] lies within 1/(2*pieces) of a grid point.
```

```
        rigorous_margin = grid_lower - derivative_upper / (2 * pieces)
        assert rigorous_margin > global_target
        print('interval:', (left, X), '; grid points:', pieces + 1)
        print('  derivative majorant:', derivative_bound)
        print('  asserted derivative upper bound:', derivative_upper)
        print('  uniform absolute tail:', tail)
        print('  smallest midpoint-selected grid lower ball:', minimum, 'at', minimizer)
        print('  every grid value strictly greater than:', grid_lower)
        print('  certified whole-interval lower bound:', rigorous_margin)
    print('CERTIFIED: g(x) > 1/20000 for every real x in [0,4].')
    print('Hence q_(3/7)(x) > Gamma(3/7)^2/20000 on [0,4].')


if __name__ == '__main__':
    main()
```